



Network Management

Having made our way through the first eight chapters of this text, we're now well aware that a network consists of *many* complex, interacting pieces of hardware and software—from the links, switches, routers, hosts, and other devices that comprise the physical components of the network to the many protocols (in both hardware and software) that control and coordinate these devices. When hundreds or thousands of such components are cobbled together by an organization to form a network, it is not surprising that components will occasionally malfunction, that network elements will be misconfigured, that network resources will be overutilized, or that network components will simply “break” (for example, a cable will be cut or a can of soda will be spilled on top of a router). The network administrator, whose job it is to keep the network “up and running,” must be able to respond to (and better yet, avoid) such mishaps. With potentially thousands of network components spread out over a wide area, the network administrator in a network operations center (NOC) clearly needs tools to help monitor, manage, and control the network. In this chapter, we'll examine the architecture, protocols, and information base used by a network administrator in this task.

9.1 What Is Network Management?

Before diving in to network management itself, let's first consider a few illustrative “real-world” non-networking scenarios in which a complex system with many interacting components must be monitored, managed, and controlled by an administrator. Electrical power-generation plants have a control room where dials, gauges, and lights monitor the status (temperature, pressure, flow) of remote valves, pipes, vessels, and other plant components. These devices allow the operator to monitor the plant's many components, and may alert the operator (with the famous flashing red warning light) when trouble is imminent. Actions are taken by the plant operator to control these components. Similarly, an airplane cockpit is instrumented to allow a pilot to monitor and control the many components that make up an airplane. In these two examples, the “administrator” *monitors* remote devices and *analyzes* their data to ensure that they are operational and operating within prescribed limits (for example, that a core meltdown of a nuclear power plant is not imminent, or that the plane is not about to run out of fuel), *reactively controls* the system by making adjustments in response to the changes within the system or its environment, and *proactively manages* the system (for example, by detecting trends or anomalous behavior, allowing action to be taken before serious problems arise). In a similar sense, the network administrator will actively monitor, manage, and control the system with which she or he is entrusted.

In the early days of networking, when computer networks were research artifacts rather than a critical infrastructure used by hundreds of millions of people a day, “network management” was unheard of. If one encountered a network problem, one might run a few pings to locate the source of the problem and then modify system settings, reboot hardware or software, or call a remote colleague to do so. (A very readable discussion of the first major “crash” of the ARPAnet on October 27, 1980, long before network management tools were available, and the efforts taken to recover from and understand the crash is [RFC 789].) As the public Internet and private intranets have grown from small networks into a large global infrastructure, the need to manage the huge number of hardware and software components within these networks more systematically has grown more important as well.

In order to motivate our study of network management, let's begin with a simple example. Figure 9.1 illustrates a small network consisting of three routers and a number of hosts and servers. Even in such a simple network, there are many scenarios in which a network administrator might benefit tremendously from having appropriate network management tools:

- *Detecting failure of an interface card at a host or a router.* With appropriate network management tools, a network entity (for example, router A) may report to the network administrator that one of its interfaces has gone down. (This is certainly preferable to a phone call to the NOC from an irate user who says the network connection is down!) A network administrator who actively monitors